

Mastering Stacks & Queues

50 Solved High-Yield Questions from GATE & PSU Exams (Last 5 Years)

Section 1: In-Depth Recent Exam Questions

Q1. GATE CSE 2026 Stack & Queue Assignment Sequence

Consider a stack SS and a queue Q , both initially empty with a capacity of 10 elements each. The elements 1, 2, 3, 4, and 5 arrive one by one in that order. When an element arrives, it is assigned either to SS (pushed) or to Q (enqueued). Once all 5 elements are stored, the output is generated by first popping all elements from SS until empty, and then dequeuing all elements from Q until empty. If the final output sequence is 4 3 1 2 5, what was the valid assignment sequence?

- (A) 1S, 2Q, 3S, 4S, 5Q
- (B) 1Q, 2Q, 3S, 4S, 5Q
- (C) 1Q, 2Q, 3Q, 4S, 5S
- (D) 1S, 2S, 3S, 4Q, 5Q

Answer: A

Explanation: Following option A: Element 1 is pushed to SS ($SS=[1]$). 2 is enqueued to Q ($Q=[2]$). 3 is pushed to SS ($SS=[1, 3]$). 4 is pushed to SS ($SS=[1, 3, 4]$). 5 is enqueued to Q ($Q=[2, 5]$). Popping all elements from SS yields 4, 3, 1. Dequeuing all elements from Q yields 2, 5. Combined output is 4 3 1 2 5.

Q2. GATE CSE 2025 Augmented Stack with Min Operation [MSQ]

We wish to augment a standard stack data structure supporting `PUSH` and `POP` with an $O(1)$ time `MIN` operation that returns a pointer to the record with the smallest key in the stack without deleting it or increasing standard operational complexities. Which of the following approach(es) can achieve this?

- (A) Keep with every record in the stack, a pointer to the record with the smallest key below it.
- (B) Keep a single pointer to the record with the smallest key in the stack.
- (C) Keep an auxiliary array in which the key values of the records are maintained in sorted order.
- (D) Keep a Min-Heap of the keys.

Answer: A

Explanation: Storing a pointer to the minimum element seen so far at each record maintains history. When a dynamic pop occurs, the state under the popped element reflects the correct previous minimum immediately in $O(1)$ time. A single pointer (B) fails on pop if the minimum is removed. Heaps/arrays (C, D) require $O(\log n)$ or $O(n)$ updates during insertions/deletions.

Q3. GATE CSE 2024 Multi-Stack Output Generation

Let S_1 and S_2 be two stacks where S_1 has a capacity of 4 elements and S_2 has a capacity of 2 elements. S_1 initially contains `[100, 200, 300, 400]` (with 400 at Top) and S_2 is empty. The only permitted operations are: `PushToS2` (pop S_1 , push to S_2), `PushToS1` (pop S_2 , push to S_1), and `GenerateOutput` (pop S_1 and output it). Which of the following output sequences can be generated?

- (A) `400, 200, 300, 100`
- (B) `400, 300, 100, 200`
- (C) `400, 300, 200, 100`
- (D) `300, 400, 200, 100`

Answer: C

Explanation: By executing `GenerateOutput` 4 times in a row, we pop from S_1 directly to the output. This yields exactly `400, 300, 200, 100`. Options like A or B require moving elements to S_2 and back, which is restricted here due to strict capacity or chronological violations.

Q4. GATE CSE 2023 Stack and Queue Interleaving Trace

Consider an array sequence $a = [1, 5, 7, 8, 9, 2]$. Stack SS and Queue Q are initially empty. The following operations are run sequentially:

1. Push all elements of a into SS (1 at bottom, 2 at top).
2. Enqueue all elements of a into Q (1 at front, 2 at rear).
3. Pop from SS . 4. Dequeue from Q . 5. Pop from SS . 6. Dequeue from Q .
7. Dequeue from Q and push that same element into SS .
8. Repeat Step 7 three times.

What is the top element of SS after executing these operations?

Answer: 2

Explanation: Initial states: $SS = [1, 5, 7, 8, 9, 2]$ (top=2), $Q = [1, 5, 7, 8, 9, 2]$ (front=1). After steps 3 to 6: 2 and 9 are popped from SS , leaving $SS = [1, 5, 7, 8]$. 1 and 5 are dequeued from Q , leaving $Q = [7, 8, 9, 2]$. Steps 7 & 8 dequeue elements 7, 8, 9, 2 in order and push them onto SS . The final pushes place 2 at the absolute top of SS .

Q5. GATE CSE 2022 In-Place Queue Reversal Cost

Consider a queue Q_1 containing four elements $[A, B, C, D]$ (A at front) and an empty queue Q_2 . The only operations allowed are Enqueue and Dequeue. What is the minimum number of Enqueue operations required on Q_1 to place the elements of Q_1 into Q_2 in reverse order (D at front) without utilizing any additional storage variables?

Answer: 6

Explanation: To pass the elements to Q_2 in reverse order (D, C, B, A), we first isolate D . We dequeue 3 elements (A, B, C) from Q_1 and immediately enqueue them back to Q_1 (3 enqueues). Then D is moved to Q_2 . Next, to isolate C , we dequeue 2 elements (A, B) and cycle them back to Q_1 (2 enqueues). C is moved to Q_2 . Finally, we cycle 1 element (A) back to Q_1 (1 enqueue). B is moved to Q_2 , then the last element A is transferred. Total enqueues on $Q_1 = 3 + 2 + 1 = 6$.

Q6. BARC OCES 2024 Stack with Primitive Reversal

An abstract data type stack supports `PUSH`, `POP`, and `REVERSE` (reverses the order of current stack elements). Which of the following is TRUE?

- (A) A queue cannot be implemented using this stack structure.
- (B) A queue can be implemented where both enqueue and dequeue take $O(1)$ amortized time.
- (C) A priority queue can be simulated in $O(1)$ time.
- (D) This structure behaves identically to a standard circular queue.

Answer: B

Explanation: By wrapping an internal `REVERSE` operation, elements can be maintained such that FIFO properties are preserved with an amortized time complexity of $O(1)$.

Q7. ISRO SCIENTIST 2023 Stack and Queue Sequential Trace

Consider the sequence of operations on an empty stack: `push(54); push(52); pop(); push(55); push(62); s = pop();`. Now consider an empty queue: `enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();`. What is the evaluated value of $s + q$?

Answer: 86

Explanation: Stack traces as: `push 54` $\rightarrow$ `[54]`; `push 52` $\rightarrow$ `[54, 52]`; `pop()` removes 52 $\rightarrow$ `[54]`; `push 55` $\rightarrow$ `[54, 55]`; `push 62` $\rightarrow$ `[54, 55, 62]`; `s = pop()` $\rightarrow$ 62. Queue traces as: `enqueue 21` $\rightarrow$ `[21]`; `enqueue 24` $\rightarrow$ `[21, 24]`; `dequeue()` removes 21 $\rightarrow$ `[24]`; `enqueue 28` $\rightarrow$ `[24, 28]`; `enqueue 32` $\rightarrow$ `[24, 28, 32]`; `q = dequeue()` $\rightarrow$ 24. Thus, $s + q = 62 + 24 = 86$.

Section 2: High-Yield Core Syllabus Database

Q8. [NIC NIELIT 2023] A single array `A[1 .. N]` implements two stacks. Stack 1 grows from 1 upward, and Stack 2 grows from `N` downward. What is the boundary condition for a "Stack Full" error?

Answer: $\text{ext}\{\text{top}\}_1 = \text{ext}\{\text{top}\}_2 - 1$. Space is optimized completely until the headers collide.

Q9. [GATE CSE 2021] A queue is implemented using a non-circular singly linked list with a `head` and `tail` pointer. If `enqueue` is implemented at the `head` and `dequeue` at the `tail`, what are the time complexities?

Answer: Enqueue is $O(1)$; Dequeue is $O(n)$ because deleting from the tail requires traversing the entire list to find and reassign the second-to-last node pointer.

Q10. [CIL MT 2022] Which data structure is best suited to optimize memory when managing a fixed-size FIFO buffer to avoid false overflow?

Answer: **Circular Queue** using an array with modulo arithmetic index wrapping.

Q11. [BARC 2023] In a circular queue implementation of size n using an array, what are the conditions for Queue Empty and Queue Full given front and rear pointers?

Answer: Empty: $\text{ext}\{\text{rear}\} == \text{ext}\{\text{front}\}$; Full: $(\text{ext}\{\text{rear}\} + 1) \% n == \text{ext}\{\text{front}\}$.

Q12. [ISRO 2021] If a stack is implemented using a singly linked list, where should **push** and **pop** operations occur to guarantee $O(1)$ performance?

Answer: At the **head** (beginning) of the linked list to bypass traversal dependencies.

Q13. [GATE CSE 2020] What is the postfix expression for the infix string: $a + b \times c - d \wedge e \wedge f$ (assuming standard precedence, $\wedge$ is right-associative)?

Answer: `abc*+def^^-`

Q14. [Vizag Steel MT 2022] Evaluate the following postfix expression using an integer stack: `8 2 3 ^ / 2 3 * + .`

Answer: **7** (Evaluation: $2^3 = 8 \rightarrow 8/8 = 1 \rightarrow 2 \times 3 = 6 \rightarrow 1+6 = 7$).

Q15. [DRDO SET 2021] Which data structure is explicitly mandatory for compilers to convert infix parsing to intermediate postfix instruction execution?

Answer: **Operator Stack**.

Q16. [AAI ATC 2023] Convert the prefix expression $- * + A B C * - D E + F G$ to its equivalent infix representation.

Answer: $((A + B) \times C) - ((D - E) \times (F + G))$.

Q17. [ISRO 2022] What is the maximum number of items present in the operator stack during the conversion of $A + (B \times C) - (D / E)$ to postfix form?

Answer: **2** (occurring when handling the $+$ and $\times$ combination within structural precedence bounds).

Q18. [GATE CSE 2021] Given an input sequence $1, 2, 3, 4, 5$, which of the following permutations cannot be generated via a stack?

Answer: `4, 1, 2, 3, 5` (Once 4 is popped, 1 cannot be accessed before elements 3 and 2 are cleared).

Q19. [NIC NIELIT 2022] A program pushes the distinct characters `a, b, c, d` onto a stack in alphabetical order, but can pop at any time. Which string cannot be an output?

Answer: `cabd`

Q20. [BARC 2022] For an input sequence of size n , the total number of distinct stack-permissible output permutations is given by which combinatorial formula?

Answer: The **Catalan Number**: $\frac{1}{n+1} \binom{2n}{n}$.

Q21. [ISRO 2023] If the push sequence is `A, B, C`, list all possible pop sequences.

Answer: `ABC, ACB, BAC, BCA, CBA` (Total of 5 distinct sequences; `CAB` is strictly impossible).

Q22. [GATE CSE 2020] What is the minimum number of stacks required to systematically implement a fully functional FIFO Queue?

Answer: 2 stacks (`SS_1` handles incoming enqueues; `SS_2` stores reversed components to drive dequeues).

Q23. [CIL MT 2023] To implement a LIFO Stack using two separate structural instances of FIFO Queues, what is the worst-case time complexity for a single `push` operation if `pop` is kept at $O(1)$?

Answer: $O(n)$ time cost per push because it requires cycling elements across queues to re-order priority layers.

Q24. [NPCIL 2022] An engineer implements a queue using two stacks `SS_1` and `SS_2`. If `SS_2` is empty during a dequeue operation, what must occur?

Answer: Pop all elements from `SS_1` and sequentially push them into `SS_2`, then pop the top element of `SS_2`.

Q25. [M fill / PSU 2021] Can a Deque (Double Ended Queue) completely mimic both a standard Stack and a standard Queue?

Answer: Yes. Restricting mutations to a single boundary acts as a stack; reading/writing on opposite boundaries satisfies queue logic.

Q26. [GATE CSE 2021] Let n insert and m ($m \leq n$) delete operations be performed on an empty queue. Let x and y be the total internal push and pop operations performed when simulating this using two stacks. Which inequality is always valid?

Answer: $2m \leq x < 2n$ and $2m \leq y \leq n + m$.

Q27. [ISRO Scientist 2022] What is the worst-case space complexity of evaluating a deeply nested postfix expression containing n operands?

Answer: $O(n)$ space bounds.

Q28. [BEL 2023] What is the time complexity to find the minimum element in a standard, un-augmented array-based stack of size n ?

Answer: $O(n)$ time complexity since it requires scanning all internal frames sequentially.

Q29. [BHAL 2022] In a circular array queue of capacity V , what is the space efficiency loss compared to a simple linear array queue?

Answer: 0 space efficiency loss. It dynamically optimizes space usage by reclaiming memory slots freed up at lower indices.

Q30. [NIC 2023] Which data structure is used natively by the Operating System to handle nested function calls and recursive subroutine allocations?

Answer: The System Call Stack.

Q31. [BARC 2024] Graph Breadth-First Search (BFS) relies on which abstract data structure to track visited adjacent vertices?

Answer: Queue.

Q32. [Vizag Steel 2021] Which tracking structure is utilized during Depth-First Search (DFS) graph traversals?

Answer: Stack.

Q33. [CIL 2023] Undo/Redo mechanisms in text editing software are built using which underlying data structure?

Answer: Two companion Stacks (one tracking historical undo edits; one tracking redo states).

Q34. [PSU Aptitude 2022] An operating system uses a printer spooler to handle print requests. Which scheduling structure handles these incoming jobs?

Answer: Queue (FIFO scheduling matrix).

Q35. [GATE CSE 2019 / Recurrent] A queue Q holds 16 numbers. An empty stack S is initialized. A loop runs: while Q is not empty, if S is empty or $\text{ext}\{\text{Top}\}(S) \leq \text{ext}\{\text{Head}\}(Q)$, dequeue Q and push to S ; otherwise pop S and enqueue back to Q . What is the maximum possible loop iterations if elements are initially sorted in descending order?

Answer: 256 iterations ($O(n^2)$ worst-case performance profile).

Q36. [ISRO 2022] What does the following function execute when passed a queue Q ?

```
void solve(Queue Q) {  
    if(isEmpty(Q)) return;  
    int x = dequeue(Q);  
    solve(Q);  
    enqueue(Q, x);  
}
```

Answer: It reverses the target queue in-place by leveraging the system execution stack implicitly.

Q37. [BARC 2023] What is the value of a function $f(S)$ over a stack SS where $f(\text{emptyset})=0$ and $f(\text{ext}\{\text{push}\}(S,i)) = \max(f(S),0)+i$? If SS contains $[2, -3, 2, -1, 2]$ from bottom to top, evaluate $f(S)$.

Answer: 4.

Q38. [NIC 2022] Can a single stack sort an input array with the help of an auxiliary variable?

Answer: No. Arbitrary sorting constraints require a minimum of two stacks or a structural recursive execution pattern.

Q39. [DRDO 2021] How many queues are required to check if a given string is a palindrome?

Answer: One Deque (Double-Ended Queue), by verifying structural token symmetry from both ends simultaneously.

Q40. [GATE CSE 2018] A priority queue is used to implement a stack. `PUSH(C)` runs `INSERT(Q, C, K)` where K is an integer key. `POP` runs `DELETEMIN(Q)`. For the stack to function correctly, the keys assigned over sequential inserts must be in:

Answer: Strictly decreasing order. The latest element gets the lowest key value, ensuring it is removed first by `DELETEMIN`.

Q41. [BEL 2022] What happens when a `POP` operation is called on a completely empty stack data structure?

Answer: Throws a system **Underflow Error**.

Q42. [CIL MT 2021] What happens when a `PUSH` operation is performed on an array-based stack whose current index matches the allocated maximum dimension?

Answer: Throws a system **Overflow Error**.

Q43. [ISRO 2023] If we implement a Queue using a circular linked list with only a single pointer, which pointer should be tracked to make both enqueue and dequeue $O(1)$?

Answer: The **REAR** pointer. `REAR->next` points directly to the front node, allowing $O(1)$ additions and deletions.

Q44. [BARC OCES 2021] Is stack allocation faster or heap allocation faster for variable runtimes in an operating system process?

Answer: **Stack allocation** is significantly faster due to its simple sequential pointer offset management model.

Q45. [NIC NIELIT 2023] Which parsing strategy uses a stack to find shift-reduce matches?

Answer: **Bottom-Up Parser** architectures.

Q46. [BHAL 2023] What is the minimum size required to store a circular queue of up to 10 elements in a standard array without a status flag variable?

Answer: **11 elements** (one index slot is left intentionally empty to mathematically differentiate a full queue from an empty one).

Q47. [DRDO 2022] If k sorted stacks are merged into a single sorted queue of size n , what is the optimal time complexity?

Answer: $O(n \log k)$ achieved by coordinating structural references via a Min-Heap layout.

Q48. [AAI 2022] Which structural principle defines a basic Queue?

Answer: **FIFO / LILO** (First-In-First-Out / Last-In-Last-Out).

Q49. [ISRO 2021] Which type of expression notation does not require structural operator precedence rules or parenthesis enclosures for evaluation?

Answer: **Postfix (Reverse Polish)** or Prefix notation.

Q50. [CIL MT 2023] Can a standard linear stack be used to evaluate an arbitrary context-free grammar?

Answer: **Yes.** This is the underlying structural foundation of a mathematical Pushdown Automaton (PDA).
